

PROJECT DOCUMENTATION

Automatic Speech Recognition Engine

1. Project Overview :-

To develop a Python-based desktop application that converts speech into text and performs various desktop and web automation tasks through voice commands. The system should make computer interaction faster, easier, and more user-friendly by allowing users to perform daily operations without using the keyboard or mouse.

2. Problem Statement

Users perform many repetitive computer tasks every day such as opening applications, visiting websites, searching information, checking the current time, or taking screenshots. Performing these tasks manually takes time and effort.

This project aims to reduce manual work by allowing users to perform these actions through simple voice commands.

3. Project Objectives

- Convert human speech into text accurately.□
 - Execute desktop automation using voice commands.□
 - Perform web automation through voice.□
 - Provide voice feedback after every command.□
 - Improve user productivity.□
 - Create a modular system that can easily support new commands in the future.□
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4. Functional Requirements:-

A. User Authentication

Purpose:

Allow only authorized users to access the voice assistant.

Functional Requirements:

- The system should allow users to log in using a username and password.
 - The system should verify the entered credentials.
 - The system should allow access only if the credentials are valid.
 - The system should display an error message for invalid credentials.
 - The system should allow users to log out.
 - The system should securely store user credentials (for a simple training project, this can be in a JSON file).
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B. Voice Input & Speech Recognition

Purpose:-

Capture the user's voice and convert it into text.

Functional Requirements:-

- The system should capture voice through the microphone.
 - The system should convert speech into text.
 - The recognized text should immediately appear on the screen. □
The system should inform the user if no speech is detected.
 - The system should handle speech recognition errors without crashing.
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C. Command Processing:-

Purpose:-

Identify the user's command and execute the correct action.

Functional Requirements:-

- The system should compare the recognized text with predefined commands.
 - The system should execute the correct action for valid commands.
 - The system should display an error message for unsupported commands.
 - The system should process only one command at a time.
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D. Desktop Automation: Purpose:-

Allow users to open desktop applications using voice commands.

Functional Requirements :-

- The system should allow users to open:
 - Calculator
 - Notepad
 - Google Chrome
 - File Explorer
 - The system should display a confirmation message after opening the application.
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E. Web Automation :-

Purpose

Open websites and perform Google searches using voice commands.

Functional Requirements

- The system should:
- Open Google.
- Open YouTube.
- Search Google using voice.
- Display an error message if the internet connection is unavailable.

Example:

User: Search Python Tutorial

Output: Google opens and searches for Python Tutorial.

F. Information Services :-

Purpose

Provide basic information to the user.

Functional Requirements :- The

system should answer:

- What is the current time?
 - What is today's date?
 - The response should be displayed on the screen and spoken aloud.
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G. Voice Response :-

Purpose :-

Allow the assistant to communicate with the user.

Functional Requirements :-

- The system should greet the user when the application starts.
- The system should speak confirmation after every successful command.
- The system should notify the user if a command fails.
- The system should say goodbye before closing the application.

➤ Input Commands & Expected Output

Input Command (User Speaks)	Expected Voice Response (System Speaks)
Hello Assistant	"Hello! How can I help you today?"
Good Morning	"Good morning! Have a great day."
Good Afternoon	"Good afternoon! How may I assist you?"
Good Evening	"Good evening! How can I help you?"
How are you?	"I'm doing great. Thank you for asking!"
What is your name?	"I am your Smart Voice Assistant."
Start Listening	"I'm listening. Please tell me your command."
Stop Listening	"Voice recognition has been stopped."

Open Google	"Opening Google."
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Open Chrome	"Opening Chrome."
Open Calculator	"Calculator has been opened."
Open Notepad	"Notepad has been opened."
Open VS Code	"Opening Visual Studio Code."
Open Downloads	"Opening Downloads folder."
Open Documents	"Opening Documents folder."
Open YouTube	"Opening YouTube."

Input Command (User Speaks)	Expected Voice Response (System Speaks)
Search Python Tutorial	"Searching Google for Python Tutorial."
Take Screenshot	"Screenshot has been captured successfully."
Increase Volume	"Volume has been increased."
Decrease Volume	"Volume has been decreased."
Mute Volume	"System volume has been muted."
Unmute Volume	"System volume has been unmuted."
What is the Time?	"The current time is 10:30 AM." <i>(Example)</i>

What is Today's Date?	"Today is 15 April 2025." <i>(Example)</i>
What Day is Today?	"Today is Tuesday." <i>(Example)</i>

Lock Computer	"Your computer is being locked."
Shutdown Computer	"Are you sure you want to shut down the computer?"
Yes	"Shutting down the computer."
No	"Shutdown operation has been cancelled."
Restart Computer	"Are you sure you want to restart the computer?"
Exit Assistant	"Thank you for using Smart Voice Assistant. Goodbye!"
Unknown Command	"Sorry, I didn't understand that command. Please try again."
No Voice Detected	"I couldn't hear anything. Please speak again."
Microphone Not Found	"No microphone detected. Please connect a microphone."
Internet Not Available	"Internet connection is unavailable. Please check your network."
Command Completed	"Your command has been completed successfully."

Command Failed	"Sorry, I couldn't complete your request."
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Example:

User: Open Google

Assistant: Google has been opened successfully.

H. Command History :-

Purpose :-

Store executed commands for future reference.

Functional Requirements

- The system should save every executed command in history.json.
 - Each record should contain:-
 - Command Name
 - Date
 - Time
 - Status
 - The system should allow users to view the command history.
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I. Voice Command Scheduling Module

Purpose:

Allow users to schedule voice commands to be executed automatically at a specified date and time.

Functional Requirements:

- The system should allow users to schedule a voice command for a specific date and time.

- The system should allow users to choose the command to be executed.
 - The system should save scheduled commands for future execution.
 - The system should automatically execute the scheduled command at the specified time.
 - The system should notify the user after successful execution.
 - The system should allow users to view all scheduled commands.
 - The system should allow users to edit scheduled commands.
 - The system should allow users to delete scheduled commands.
 - The system should display an error message for invalid date or time.
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J. Voice Calculator Module

Purpose:

Perform mathematical calculations using voice commands.

Functional Requirements:

- The system should accept mathematical expressions through voice.
 - The system should convert speech into text.
 - The system should validate the mathematical expression.
 - The system should evaluate the mathematical expression.
 - The system should display the calculated result on the screen.
 - The system should speak the calculated result.
 - The system should maintain a history of previous calculations.
 - The system should display an error message for invalid expressions.
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K. Dashboard Module

Purpose:

Provide a centralized interface to monitor and manage the voice assistant.

Functional Requirements:

- The system should display the current user information.
 - The system should display the last executed voice command.
 - The system should display the command execution history.
 - The system should display the total number of executed commands.
 - The system should display scheduled commands.
 - The system should display the current system status (Microphone and Internet).
 - The system should provide quick access to all available modules.
 - The system should refresh the displayed information automatically.
 - The system should display notifications for successful and failed operations.
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L. Email Sending Through Voice Command Module

Purpose:

Allow users to send emails using voice commands.

Functional Requirements:

- The system should allow users to specify the recipient's email address through voice.
- The system should allow users to specify the email subject through voice.
- The system should allow users to dictate the email message through voice.
- The system should convert the spoken content into text.
- The system should allow users to attach PDF, Word, Excel, or image files.
- The system should ask for confirmation before sending the email.
- The system should send the email to the specified recipient.
- The system should display a success message after sending the email.

- The system should display an error message if the email cannot be sent.
 - The system should maintain a history of sent emails.
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5. JSON Files Used

File Name	Purpose
commands.json	Stores predefined voice commands
history.json	Stores command execution history
users.json	Stores usernames and passwords

6. Error Handling Module :-

Requirements:

- Handle microphone not found.□
 - Handle internet unavailable.□
 - Handle unsupported commands.□
 - Handle application not installed.□
 - Prevent the program from crashing.□
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7.Project Scope :-

➤ In Scope

- Speech Recognition□
- Desktop Automation□
- Web Automation□
- Voice Feedback□
- Command History using JSON□

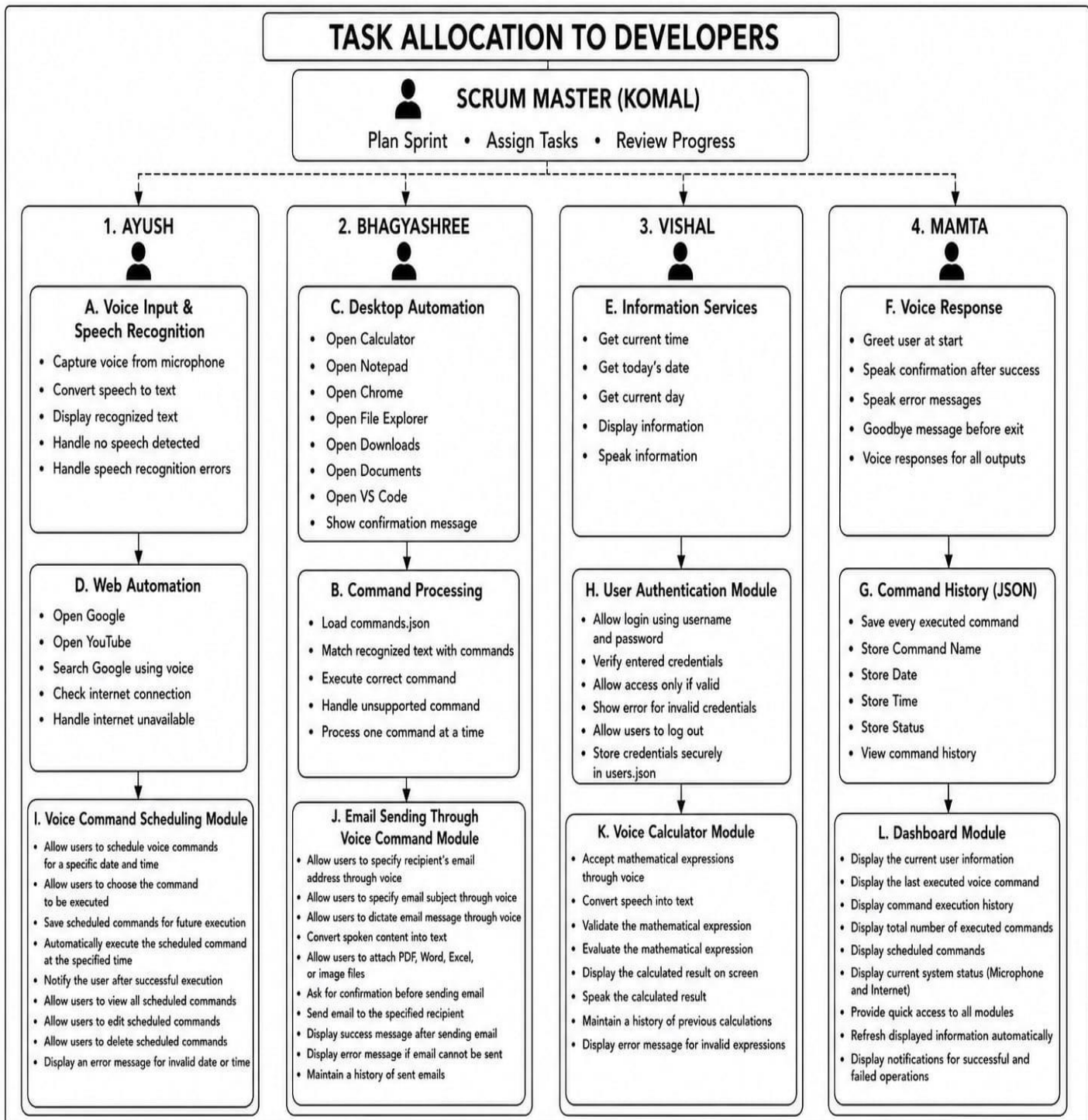
➤ Out of Scope/Future Scope

- Mobile Application□
 - AI Chatbot□
 - Cloud Deployment□
 - Custom Voice Commands□
 - Advanced NLP□
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8.User Stories :-

- As a user, I want to open applications using voice commands.
 - As a user, I want to search Google without typing. □ As a user, I want to open websites using voice.
 - As a user, I want the assistant to speak responses.
 - As a user, I want my command history to be saved.
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9. Task Allocation



10.Product Backlog Grooming

Product Backlog ID	User Story	Priority	Estimated Effort
PB-01	As a user, I want to log in securely using a username and password before accessing the voice assistant.	High	3 Hrs
PB-02	As a user, I want the system to capture my voice through the microphone and convert it into text accurately.	High	4 Hrs
PB-03	As a user, I want the assistant to recognize my spoken command and execute the correct action.	High	4 Hrs
PB-04	As a user, I want to open desktop applications (Calculator, Notepad, Chrome, File Explorer, VS Code) using voice commands.	High	4 Hrs
PB-05	As a user, I want to open websites and search Google using voice commands.	High	3 Hrs
PB-06	As a user, I want to get information like the current time, date, and day through voice responses.	Medium	2 Hrs
PB-07	As a user, I want the assistant to respond with voice feedback after every successful or failed command.	High	3 Hrs
PB-08	As a user, I want to control system functions such as taking screenshots, adjusting volume, locking, restarting, and shutting down the computer using voice commands.	High	5 Hrs
PB-09	As a user, I want every executed command to be stored in history.json with its date, time, and status.	Medium	3 Hrs
PB-10	As a user, I want the system to handle microphone, internet, and unsupported command errors without crashing.	High	3 Hrs
PB-11	As a user, I want all modules integrated into one menu-driven voice assistant application.	High	4 Hrs
PB-12	Testing, bug fixing, integration testing, and final project documentation.	High	5 Hrs

10.WorkFlow

Start



System Initialization

→ Load commands.json and history.json

→ Check microphone availability

→ Initialize required modules



Welcome Message

→ Greet the user through voice response



Listen for Voice Command

→ Capture user's voice through the microphone



Speech Recognition

→ Convert speech into text

→ Display the recognized text on the screen



Process Command

→ Compare the recognized text with predefined commands

→ Validate the command

→ Identify the requested operation



Execute Command (Based on User Input)

1. Desktop Automation → Open Calculator

→ Open Notepad

→ Open Google Chrome

- Open File Explorer
- Open Downloads Folder
- Open Documents Folder
- Open VS Code
- Display confirmation message

2. Web Automation

- Open Google
- Open YouTube
- Search Google using voice command
- Check internet connection
- Display confirmation message

3. Information Services → Get current time

- Get today's date
- Get current day
- Display information on screen
- Speak the requested information

4. System Control

- Capture Screenshot
- Increase Volume
- Decrease Volume
- Mute / Unmute Volume
- Lock Computer
- Restart Computer (with confirmation)
- Shutdown Computer (with confirmation)

5. User Authentication

- Verify username and password
- Allow access if credentials are valid

→ Display error message for invalid credentials

→ Allow user to log out



Voice Response

→ Speak success message

→ Speak error message if command fails

→ Provide requested information through voice



Save Command History

→ Store command name

→ Store execution date →

Store execution time

→ Store execution status in history.json



Wait for Next Command

→ Continue listening for the next voice command



Exit Assistant

→ Speak goodbye message

→ Close the application

9.Acceptance Criteria :-

- **The project will be considered successful if: ☐ Speech is recognized correctly.**
- **Supported commands execute successfully.**
- **Applications and websites open correctly.**
- **Voice responses are provided.**
- **Command history is stored in history.json.**
- **The application handles invalid commands without crashing.**